Code No: 138GK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
MICROPROCESSORS AND MICROCONTROLLERS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) With a neat block diagram, explain the architecture of 8086 microprocessor.
b) What are assembler directives? List some of the assembler directives and explain? [10+5]

2.a) Explain the operation of 8086 in its minimum mode with timing diagram.
b) Write an assembly language program to move a block of 10KB of data from 35000H location to 25000H location. [10+5]

3.a) Explain in detail about the memory organization of 8051 memory controller.
b) Draw the 8051 port diagram, explain various I/O ports of 8051. [8+7]

4.a) Explain the TMOD function register and its timer modes of operations.
b) Explain the addressing modes of 8051 with an example for each. [8+7]

5.a) Draw the block schematic of DAC interfaced to 8051 at port P1 and write an 8051 program to generate sine wave.
b) Explain the different steps to receive data serially using 8051. [10+5]

6.a) How to interface an LCD display with microcontroller? Explain how to display a character using LCD display.
b) What is serial communication? How is this achieved with 8051 using RS232 standards? [8+7]

7.a) With an example each, explain the Thumb data processing instructions.
b) Draw and explain current program status register of ARM processor. [10+5]

8.a) With the help of a neat block diagram, explain the architecture of ARM cortex M3 processor.
b) Explain the register organization of cortex M3. [10+5]

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